ALGEBRA 1 · SOLVING LINEAR EQUATIONS

Equation Balance Scale

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Equation Balance Scale** on Mr. K's Classroom: Algebra 1 → Solving Linear Equations → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. On the scale for 2x + 3 = 11, what will happen if you subtract 3 from the left side only?

   The scale tips. The left side gets lighter, so the two sides are no longer equal.

2. To solve 2x + 3 = 11, which operation would you do first? Why?

   Subtract 3 from both sides first, to undo the + 3 before undoing the × 2.

2 Explore use the tool and record what you see

Solve on the scale

Use only operations on both sides. List your steps, then check your answer in the original equation.

Equation	Your steps	Solution	Check
A: 2x + 3 = 11	**subtract 3; divide by 2**	**x = 4**	**2(4) + 3 = 11 ✓**
B: 3x – 5 = 10	**add 5; divide by 3**	**x = 5**	**3(5) – 5 = 10 ✓**
C: 5x + 2 = 2x + 14	**subtract 2x; subtract 2; divide by 3**	**x = 4**	**22 = 22 ✓**
D: 6x – 4 = 2x + 8	**subtract 2x; add 4; divide by 4**	**x = 3**	**14 = 14 ✓**

When the x-terms cancel

In equations E and F, subtract the x-terms from both sides.

Equation	What is left	True or false?	Number of solutions
E: 2x + 10 = 2x + 10	**10 = 10**	**true**	**infinitely many**
F: 3x + 4 = 3x – 2	**4 = –2**	**false**	**none**

3 Write the rule fill in each blank from what you observed

1. To keep an equation balanced, whatever you do to one side you must also do to **the other side**.
2. To undo addition, **subtract**. To undo multiplication, **divide**.
3. If the x-terms cancel and what is left is true (like 10 = 10), the equation has **infinitely many solutions**.
4. If the x-terms cancel and what is left is false (like 4 = –2), the equation has **no solution**.

4 Check no tool: show your work

1. Solve 4x – 7 = 13.

   x = 5

2. Solve $7x + 3 = 4x + 15$.

   x = 4

3. Solve $\frac{x}{3} + 2 = 6$.

   x = 12

4. How many solutions does $2(x + 5) = 2x + 10$ have?

   Infinitely many

5. How many solutions does $5x - 1 = 5x + 6$ have?

   None

5 Reflect

Why do you have to do the same thing to both sides of an equation?

An equation says both sides are equal. Changing only one side breaks the balance, so the new equation would no longer have the same solution.